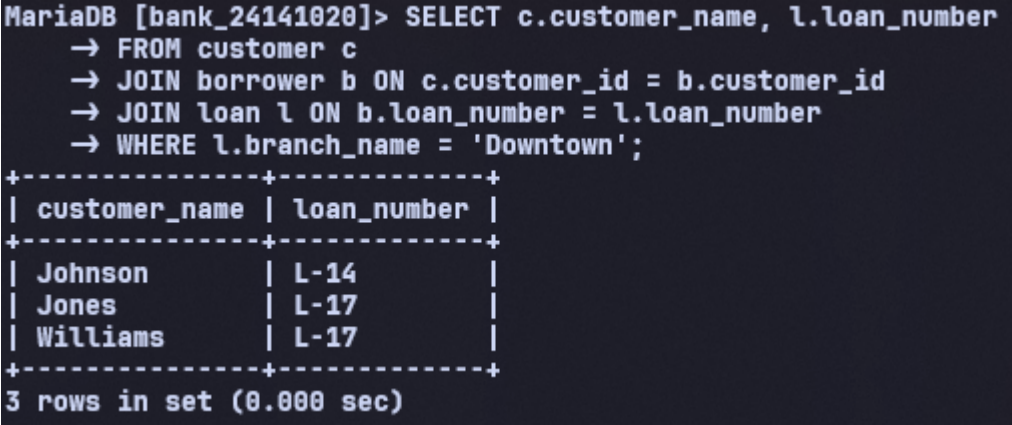


CSE370 : Database Systems

Assignment 03 | Spring 2026

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No 1 Query (as Plain Text)	SELECT c.customer_name, l.loan_number FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE l.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	 <pre>MariaDB [bank_24141020]> SELECT c.customer_name, l.loan_number → FROM customer c → JOIN borrower b ON c.customer_id = b.customer_id → JOIN loan l ON b.loan_number = l.loan_number → WHERE l.branch_name = 'Downtown'; +-----+-----+ customer_name loan_number +-----+-----+ Johnson L-14 Jones L-17 Williams L-17 +-----+-----+ 3 rows in set (0.000 sec)</pre>
No 2 Query (as Plain Text)	SELECT DISTINCT c.customer_name FROM customer c JOIN borrower b ON c.customer_id = b.customer_id;

No 2 SS (of Query & Output in Shell)	<pre> MariaDB [bank_24141020]> SELECT DISTINCT c.customer_name → FROM customer c → JOIN borrower b ON c.customer_id = b.customer_id; +-----+ customer_name +-----+ Jones Smith Hayes Curry Williams Adams Johnson +-----+ 7 rows in set (0.000 sec) </pre>
No 3 Query (as Plain Text)	<pre> SELECT c.customer_name FROM customer c LEFT JOIN borrower b ON c.customer_id = b.customer_id WHERE b.loan_number IS NULL; </pre>

No 3 SS
(of Query & Output
in Shell)

```
MariaDB [bank_24141020]> SELECT c.customer_name
→ FROM customer c
→ LEFT JOIN borrower b ON c.customer_id = b.customer_id
→ WHERE b.loan_number IS NULL;

+-----+
| customer_name |
+-----+
| Lindsay      |
| Turner       |
| Glenn        |
| Brooks       |
| Green        |
+-----+
5 rows in set (0.000 sec)
```

No 4 Query
(as Plain Text)

```
SELECT c1.customer_name AS Customer1,
       c2.customer_name AS Customer2,
       c1.customer_city AS City
FROM customer c1
JOIN customer c2
ON c1.customer_city = c2.customer_city
AND c1.customer_id < c2.customer_id;
```

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [bank_24141020]> SELECT c1.customer_name AS Customer1,  
→      c2.customer_name AS Customer2,  
→      c1.customer_city AS City  
→ FROM customer c1  
→ JOIN customer c2  
→ ON c1.customer_city = c2.customer_city  
→ AND c1.customer_id < c2.customer_id;  
+-----+-----+-----+  
| Customer1 | Customer2 | City |  
+-----+-----+-----+  
| Jones     | Hayes     | Harrison |  
| Smith     | Curry     | Rye      |  
| Lindsay   | Adams     | Pittsfield |  
| Turner    | Green     | Stamford |  
+-----+-----+-----+  
4 rows in set (0.000 sec)
```

No 5 Query
(as Plain Text)

```
SELECT branch_name,  
ROUND(SUM(balance * 0.04),2) AS Total_Interest  
FROM account  
GROUP BY branch_name;
```

No 5 SS
(of Query & Output
in Shell)

```
MariaDB [bank_24141020]> SELECT branch_name,  
→ ROUND(SUM(balance * 0.04),2) AS Total_Interest  
→ FROM account  
→ GROUP BY branch_name;  
+-----+-----+  
| branch_name | Total_Interest |  
+-----+-----+  
| Brighton    | 66.00          |  
| Downtown    | 20.00          |  
| Mianus       | 28.00          |  
| Perryridge  | 16.00          |  
| Redwood     | 28.00          |  
| Round Hill  | 14.00          |  
+-----+-----+  
6 rows in set (0.000 sec)
```

No 6 Query
(as Plain Text)

```
SELECT b.branch_city, a.account_number  
FROM account a  
JOIN branch b ON a.branch_name = b.branch_name  
WHERE a.balance = (  
    SELECT MAX(a2.balance)  
    FROM account a2  
    JOIN branch b2 ON a2.branch_name = b2.branch_name  
    WHERE b2.branch_city = b.branch_city  
);
```

No 6 SS
(of Query & Output
in Shell)

```
MariaDB [bank_24141020]> SELECT b.branch_city, a.account_number
→ FROM account a
→ JOIN branch b ON a.branch_name = b.branch_name
→ WHERE a.balance = (
→     SELECT MAX(a2.balance)
→     FROM account a2
→     JOIN branch b2 ON a2.branch_name = b2.branch_name
→     WHERE b2.branch_city = b.branch_city
→ );
```

branch_city	account_number
Brooklyn	A-201
Horseneck	A-215
Palo Alto	A-222

```
3 rows in set (0.001 sec)
```

No 7 Query
(as Plain Text)

```
SELECT l.loan_number, l.amount, c.customer_name
FROM loan l
JOIN borrower b ON l.loan_number = b.loan_number
JOIN customer c ON b.customer_id = c.customer_id
ORDER BY l.amount DESC, l.loan_number ASC
LIMIT 5;
```

No 7 SS
(of Query & Output
in Shell)

```
MariaDB [bank_24141020]> SELECT l.loan_number, l.amount, c.customer_name
→ FROM loan l
→ JOIN borrower b ON l.loan_number = b.loan_number
→ JOIN customer c ON b.customer_id = c.customer_id
→ ORDER BY l.amount DESC, l.loan_number ASC
→ LIMIT 5;

+-----+-----+-----+
| loan_number | amount | customer_name |
+-----+-----+-----+
| L-23       | 2000   | Smith         |
| L-14       | 1500   | Johnson       |
| L-15       | 1500   | Hayes         |
| L-16       | 1300   | Adams         |
| L-17       | 1000   | Jones         |
+-----+-----+-----+
5 rows in set (0.000 sec)
```

No 8 Query
(as Plain Text)

```
SELECT DISTINCT c.customer_name
FROM customer c
JOIN depositor d ON c.customer_id = d.customer_id
JOIN account a ON d.account_number = a.account_number
JOIN borrower b ON c.customer_id = b.customer_id
JOIN loan l ON b.loan_number = l.loan_number
WHERE a.branch_name = 'Perryridge'
AND l.branch_name = 'Perryridge';
```

No 8 SS
(of Query & Output in
shell)

```
MariaDB [bank_24141020]> SELECT DISTINCT c.customer_name
→ FROM customer c
→ JOIN depositor d ON c.customer_id = d.customer_id
→ JOIN account a ON d.account_number = a.account_number
→ JOIN borrower b ON c.customer_id = b.customer_id
→ JOIN loan l ON b.loan_number = l.loan_number
→ WHERE a.branch_name = 'Perryridge'
→ AND l.branch_name = 'Perryridge';

+-----+
| customer_name |
+-----+
| Hayes         |
+-----+
1 row in set (0.000 sec)
```

No 9 Query
(as Plain Text)

```
SELECT c.customer_name, SUM(l.amount) AS total_loan
FROM customer c
JOIN borrower b ON c.customer_id = b.customer_id
JOIN loan l ON b.loan_number = l.loan_number
GROUP BY c.customer_name
HAVING COUNT(l.loan_number) >= 2;
```


No 9 SS
(of Query & Output in
shell)

```
MariaDB [bank_24141020]> SELECT c.customer_name, SUM(l.amount) AS total_loan
→ FROM customer c
→ JOIN borrower b ON c.customer_id = b.customer_id
→ JOIN loan l ON b.loan_number = l.loan_number
→ GROUP BY c.customer_name
→ HAVING COUNT(l.loan_number) ≥ 2;
+-----+-----+
| customer_name | total_loan |
+-----+-----+
| Smith        |         2900 |
+-----+-----+
1 row in set (0.000 sec)
```

No 10 Query
(as Plain Text)

```
SELECT d.customer_id, SUM(a.balance) AS total_balance
FROM depositor d
JOIN account a ON d.account_number = a.account_number
GROUP BY d.customer_id
HAVING COUNT(d.account_number) > 1;
```

No 10 SS
(of Query & Output in
shell)

```
MariaDB [bank_24141020]> SELECT d.customer_id, SUM(a.balance) AS total_balance
→ FROM depositor d
→ JOIN account a ON d.account_number = a.account_number
→ GROUP BY d.customer_id
→ HAVING COUNT(d.account_number) > 1;
+-----+-----+
| customer_id | total_balance |
+-----+-----+
| C-226      |         1400 |
+-----+-----+
1 row in set (0.000 sec)
```